Early-life mortality rates in Brazil and progress toward meeting related sustainable development goals: a nationwide population study

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Summary

Background Early-life mortality rates have significantly declined since 2000, and sustaining this progress could save over 11 million children's lives worldwide. Achieving uniform reductions in early-life mortality remains a critical public health challenge in Brazil. This study analyzed disaggregated trends in early-life mortality across Brazilian regions and assessed each region's progress toward meeting the Sustainable Development Goals (SDGs) and national targets for Brazil.

Methods This is a population-based analytical and ecological study. We analyzed data from official government information systems for Brazilian regions and states over five consecutive five-year periods between 1998 and 2022. The SDG targets considered were: an under-five mortality rate (U5MR) of 25 per 1000 live births, a neonatal mortality rate (NMR) of 12.0 per 1000 live births, and an infant mortality rate (IMR) of 15.7 per 1000 live births. Brazil's adjusted national targets were: U5MR of 8.3, NMR of 5.3, and IMR of 7.7 per 1000 live births, respectively. We calculated the absolute annual change (AAC) in mortality rates for each five-year interval to assess trends and infer the likelihood of meeting the 2030 targets.

Findings Between 1998 and 2022, U5MR in Brazil declined by 54%, from 32.5 to 15.1 deaths per 1000 live births. The national average met the SDG targets for all age strata, but no state achieved the stricter IPEA targets. From 2018 to 2022, under-five and infant mortality rates showed annual increases (AAC = +0.22 and + 0.10, respectively), while neonatal mortality remained stable (AAC = −0.02). Regional disparities persisted. In 2022, Santa Catarina (South region) had the lowest U5MR (11.5 deaths per 1000 live births), while Roraima (North region) had the highest (24.6 deaths per 1000 live births). Although most states are projected to meet the SDG targets for U5MR by 2030, none are on track to meet the IPEA goals.

Interpretation Despite national progress in meeting SDG targets for early-life mortality, recent increases in under-five and infant deaths and enduring regional inequalities threaten future gains. Sustained improvements will require focused, region-specific strategies to address the social and health determinants driving these disparities.

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Keywords: Under-five mortality; Neonatal mortality; Infant mortality; Sustainable development goals (SDGs); Brazil

Introduction

The Millennium Development Goals were a series of targets established by the global community to combat poverty, hunger, environmental degradation and improve global health. The Millennium Development Goals number 4 aimed to reduce the under-five

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Research in context

Evidence before this study

Despite significant global progress in reducing early-life mortality rates, the patterns of decline vary greatly between regions. For instance, the under-five mortality rate (U5MR) ranges from 5 to 71 per 1000 live births, globally. International organizations, such as the World Health Organization and UNICEF, have emphasized the importance of subnational analyses to better understand disparities and to guide strategies for achieving the Sustainable Development Goals (SDGs). In a recent study, Subramanian (2024) explored disaggregated trends in early-life mortality across Indian states and projected their progress towards the SDG targets. However, similar comprehensive studies for Brazil are lacking. To address this gap, we conducted a systematic search on PubMed for studies examining trends in early-life mortality rates and SDG targets in Brazil, using the terms: "Sustainable Development Goals" AND "Under-five Mortality" OR "Neonatal Mortality" OR "Infant Mortality" AND "Brazil". This search yielded 13 relevant articles, but only one provided a detailed analysis of health-related SDGs across Brazilian states. However, this study did not specifically focus on early-life mortality rates. Other studies focused on specific Brazilian cities, highlighting localized trends rather than offering a national perspective. To our knowledge, no prior research has examined the decreasing patterns of early-life mortality rates across all Brazilian states, nor has any study assessed their progress in meeting SDGs targets at a disaggregated, state-by-state level.

Added value of this study

This is the first study to describe disaggregated patterns of early-life mortality across Brazilian states over a 25-year period, providing a comprehensive and detailed view of regional variations and progress towards SDG targets. It offers a detailed examination of regional variations in mortality reduction and evaluates the progress of each state toward achieving both the global SDG targets and Brazil's more stringent adjusted targets. The study also identifies states that may achieve these goals by 2030, especially in the neonatal mortality category.

Implications of all the available evidence

Understanding geographical disparities in early-life mortality is crucial for developing targeted health interventions and policies that can accelerate progress toward SDG targets. By identifying regions where progress has lagged, this study provides a foundation for addressing inequities in child survival rates and informs strategies to improve early-life outcomes across Brazil. Policymakers can use these insights to prioritize resources and tailor interventions to the specific needs of the most vulnerable populations, ensuring that the benefits of improved healthcare reach all Brazilian children.

mortality rate (U5MR) by two-thirds between 1990 and 2015.[1] Although this target was not fully achieved, significant progress was made, laying a foundation for a new set of targets, the Sustainable Development Goals (SDGs), outlined by the United Nations (UN) in 2015. One of the targets (SDG 3) focused on ensuring healthy lives and promoting well-being for all at all ages, and included the goals of ending the preventable deaths of newborns and children under five, achieving a U5MR lower than 25 deaths per 1000 live births, and a neonatal mortality rate (NMR) below 12 deaths per 1000 live births.[2]

By 2022, the global USMR had decreased by 51% (to 37 deaths per 1000 live births) and NMR by 44% (to 17 deaths per 1000 live births) compared to 2000. In Latin America, U5MR decreased by 52%, and NMR by 54% over the same period. Sub-Saharan Africa had the highest rates (U5MR = 71/1000; NMR = 27/1000), while North America and Europe had the lowest (U5MR = 5/1000; NMR = 3/1000).[2] Achieving the SDG targets globally could save over 11 million children's lives by 2030.[3]

Between 2000 and 2015, Brazil significantly reduced early-life mortality rates, surpassing the SDG targets ahead of the 2030 deadline.[4] Consequently, the Institute for Applied Economic Research (*Instituto de Pesquisa Econômica Aplicada* [IPEA], in Portuguese), a Brazilian governmental institution that conducts research and provides technical advice on public policy and socio-economic issues, set new targets for 2030, aiming for an NMR of 5.3 per 1000 live births and a U5MR of 8.3 per 1000 live births.[5]

Brazil has an estimated population of 212 million inhabitants, and comprises 26 states and one federal district.[6] It is conventionally divided into five geographical regions (North, Northeast, Southeast, South, and Central-West). Historically, the best development rates have been in the South and Southeast regions. Seven of the ten Brazilian states with the highest human development index are in these regions, with the other three in the Central-West. Conversely, the ten states with the lowest human development index are in the North or Northeast.[7] Over the last three decades, Brazil has experienced significant economic growth, alongside reductions in illiteracy and poverty, positioning it among upper-middle-income countries, according to the World Bank's classification.[8,9]

Brazil has one of the largest public health systems in the world, with most of the population relying on it. The Unified Health System (*Sistema Único de Saúde* [SUS], in Portuguese) is funded and provided at three levels (federal, state and municipal), each one with specific responsibilities according to the complexity of care required.[8] Despite the significant improvements in health service coverage and reductions in mortality rates, regional differences persist, and uniformly reducing early-life mortality remains a critical public health challenge.[2,10–13]

Understanding regional patterns is essential for developing targeted health interventions. While comprehensive studies on neonatal and child mortality have been conducted in other countries, such as the work by Subramanian et al. (2024) in India,[14] similar analyses for Brazil are lacking. This study aims to explore the disaggregated patterns of decreased early-life mortality across Brazilian regions, and analyze each region's progress toward achieving both the SDG and IPEA targets.

Methods

Study design, population, and data source

This is a population-based analytical and ecological study whose aim is to: identify regional patterns in the reduction of early-life mortality across Brazil between 1998 and 2022; and assess each region's progress toward achieving the global (SDG 3/UN) and national adjusted (IPEA).

The analysis covers the period from 1998 (as data from earlier years were less reliable in several states due to the uneven implementation of the information systems) to 2022, the most recent year with fully consolidated data at the time we extracted the data. Data were grouped into five-year periods to ensure more stable estimates and facilitate temporal comparisons, as adopted in similar national studies: 1998–2002, 2003–2007, 2008–2012, 2013–2017, and 2018–2022.

Data were obtained from two publicly available databases.[4] Data on childhood deaths between the ages 0 to 5 were obtained from the Mortality Information System. The Mortality Information System is a comprehensive national database managed by the Ministry of Health. It records and monitors mortality data nationwide, including causes of death and demographic details. This data is vital for public health planning, guiding policy decisions, and health interventions. The system covers all regions, enabling tracking of mortality trends, assessing the impact of health initiatives, and identifying areas for improvement. The number of live births for each period was obtained from the Live Birth Information System (SINASC, in Portuguese). SINASC is a nationwide database managed by the Ministry of Health, which collects detailed data on live births across the entire country. All data were related to the mother's place of residence (state), and all under-five deaths registered in the Mortality Information System for all Brazilian states were included in the study.

Statistical analysis and data interpretation

We computed the following mortality rates: a) NMR–deaths among babies during the first twenty-eight days of life per 1000 live births; b) Infant mortality rate (IMR)—deaths among children less than one year old per 1000 live births; and c) U5MR–deaths among infants aged 0–59 months per 1000 live births. Mortality rates were calculated by Brazilian states, grouped into geographic regions (North, Northeast, Southeast, South, and Central-West) (Supplementary Fig. S1), in each study period.

Using the mortality rate data, we estimated the absolute annual change (AAC) for each rate and the change over the study period. $AAC = (R2 - R1)/T$, where R1 is the mortality rate in the initial year, R2 is the mortality rate in the final year, and T is the time elapsed between the evaluated years. The AAC values provide insight into trends over time: an AAC with negative values indicates a decrease in the analyzed rates, an AAC with positive values indicates an increase, and an AAC with values close to zero denotes a stable trend.

For the analysis of mortality rates status, we used the SDG targets of 25/1000 live births for U5MR, 12/1000 live births for NMR, and 15.7/1000 live births for IMR.[1] We also considered the more stringent targets defined by the IPEA targets of: 8.3/1000 live births for U5MR, 5.3/1000 live births for NMR, and 7.7/1000 live births for IMR.[5]

The required AACs to meet the SDG and IPEA targets by 2030 were calculated and compared with the AAC for the last assessed period (2018–2022). A positive $AAC_{2022–2030}$ indicates that the rate was within the target in 2022, and only an $AAC_{2018–2022}$ higher than the $AAC_{2022–2030}$ would imply that the target would not be met by 2030. Conversely, a negative $AAC_{2022–2030}$ suggests that the rate was not within the target in 2022, and only an $AAC_{2018–2022}$ equal to or lower than the $AAC_{2022–2030}$ would result in achieving the target by 2030. Given that the IPEA targets are more stringent than the SDG targets, the following scenarios were identified:

1. Met the IPEA target by 2022 and will continue to meet it in 2030.
2. Met the SDG target by 2022 and will continue to meet it in 2030; did not meet the IPEA target by 2022 but will meet it in 2030.
3. Met the SDG target by 2022 and will continue to meet it in 2030; did not meet the IPEA target by 2022 and will not meet it in 2030.
4. Met the SDG target by 2022, but will not continue to meet it in 2030.
5. Did not meet the SDG target by 2022 and will not meet it in 2030.

We categorized U5MR in the following age periods: a) early neonatal period—the first six days of life; b) late neonatal—from seven to twenty-eight days of life; c) post-neonatal—from one to eleven months of life; d) child period—from one to four years old. The proportions of each age group within the Under-5 mortality were displayed in a bar graph, showing the national average and regional data for the evaluated periods.

Data were analyzed using R software version 4.3.1 for Windows (R Core Team 2023).

Ethical considerations

As data in the Mortality Information System and Live Birth Information System are deidentified and publicly available, institutional review board approval and informed consent were not required.

Role of funding source

The study had no funder.

Results

In Brazil, there was a significant decrease in mortality among children over the 25 years studied. The risk of a child dying before reaching the age of five was reduced by 54%, from 32.5 per 1000 live births in 1998 to 15.1 per 1000 live births in 2022. Fig. 1 presents boxplots illustrating the evolution of mortality rates for the three age strata (neonatal, infant, and under-five) across the five study periods. A progressive reduction in the median for each mortality rate was observed in the initial years, followed by a period of stabilization and a slight increase in the final years. The rates among states exhibited lower variability in the neonatal group and wider variability in the infant and under-five groups. However, the range in rates decreased over time. In 2022, for example, Santa Catarina (Southeast region) recorded the lowest mortality rates across all three age groups assessed: NMR of 6.83, IMR of 9.8, and U5MR of 11.5 deaths per 1000 live births. In contrast, the highest NMR was observed in Sergipe (Northeast region), at 11.9. For both IMR and U5MR, the highest rates were found in Roraima (North region), at 18.8 and 24.6, respectively. These findings underscore persistent regional inequalities in child mortality, with the Southeast region performing best and the North and Northeast regions facing the greatest challenges.

The SDG target (blue line in Fig. 1) was achieved concerning the national average for all age strata in the first decade of the millennium. For U5MR, all states were under the SDG target from the second period of study (2003–2007), while for NMR, all states are under the SDG target in the last years. However, for IMR, a few states (four) are still off track until the final years. On the other hand, the adjusted goals (IPEA target—green line in Fig. 1) were not met in any state, and many states fell far short of the target.

Tables 1–3 show mortality rates for each age group (under-5, infant, and neonatal, respectively) across the five studied periods, segmented by regions and states, including the AAC. Fig. 2 shows the most recent mortality rates (2022), and the AAC from 2018 to 2022, with the AAC colored to indicate whether the SDG and IPEA targets were met or not.

In respect of the national average, U5MR showed the highest annual decrease in the overall period of study ($AAC_{1998–2022}$ = −0.73), while the NMR exhibited the smallest decrease ($AAC_{1998–2022}$ = −0.25). In the most recent evaluation period (2018–2022), both the U5MR ($AAC_{2018–2022}$ = +0.22) and IMR ($AAC_{2018–2022}$ = +0.10) experienced annual increases, whereas the NMR remained stable with an $AAC_{2018–2022}$ of −0.02.

Concerning the Brazilian states, the highest decrease for U5MR was observed in Paraiba (Northeast region) ($AAC_{1998–2022}$ = −1.83); for IMR was observed in

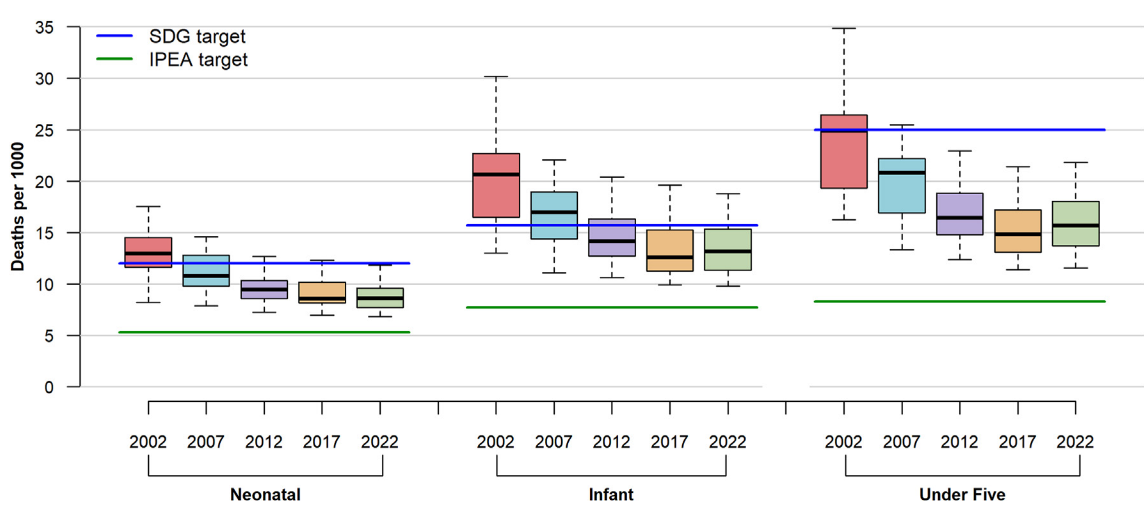

Fig. 1: Time evolution of the statistical distribution of neonatal, infant, and under-5 mortality rates across Brazilian states, 1998–2022. Legends/Abbreviations: SDG = Sustainable Development Goals, IPEA = Instituto de Pesquisa Econômica Aplicada (Applied Economic Research).

Region/State	Under-five mortality rate per 1000 live births (AAC)					
	1998–2002	2003–2007	2008–2012	2013–2017	2018–2022	1998–2022
Brazil	**32.46–22.51 (−2.49)**	**22.26–18.40 (−0.97)**	**17.62–15.64 (−0.49)**	**15.60–14.41 (−0.30)**	**14.17–15.04 (0.22)**	**32.46–15.04 (−0.73)**
North Region	**29.55–26.25 (−0.82)**	**25.95–22.11 (−0.96)**	**21.23–19.88 (−0.34)**	**19.85–18.32 (−0.38)**	**18.07–18.52 (0.11)**	**29.55–18.52 (−0.46)**
Rondônia	28.94–24.90 (−1.01)	25.48–21.92 (−0.89)	19.48–15.99 (−0.87)	16.79–14.73 (−0.52)	14.31–16.18 (0.47)	28.94–16.18 (−0.53)
Acre	40.63–25.00 (−3.91)	23.96–25.21 (0.31)	21.12–19.64 (−0.37)	19.91–16.63 (−0.82)	19.46–20.64 (0.30)	40.63–20.64 (−0.83)
Amazonas	39.77–28.39 (−2.85)	26.42–21.03 (−1.35)	20.55–20.26 (−0.07)	20.89–19.70 (−0.30)	19.34–19.46 (0.03)	39.77–19.46 (−0.85)
Roraima	31.38–16.46 (−3.73)	19.29–20.77 (0.37)	19.90–21.13 (0.31)	23.86–21.39 (−0.62)	23.98–24.60 (0.15)	31.38–24.60 (−0.28)
Pará	26.98–26.56 (−0.10)	27.12–22.36 (−1.19)	21.88–20.50 (−0.35)	19.63–18.20 (−0.36)	17.57–17.76 (0.05)	26.98–17.76 (−0.38)
Amapá	22.98–28.39 (1.35)	23.57–22.74 (−0.21)	25.49–22.96 (−0.63)	22.98–23.83 (0.21)	21.05–21.81 (0.19)	22.98–21.81 (−0.05)
Tocantins	16.59–23.41 (1.70)	23.80–22.03 (−0.44)	19.25–17.05 (−0.55)	17.33–14.84 (−0.62)	15.27–15.48 (0.05)	16.59–15.48 (−0.05)
Northeast Region	**38.48–27.79 (−2.67)**	**27.14–21.47 (−1.42)**	**20.08–17.38 (−0.68)**	**17.81–16.27 (−0.38)**	**15.68–16.69 (0.25)**	**38.48–16.69 (−0.91)**
Maranhão	15.16–25.23 (2.52)	23.73–20.83 (−0.73)	19.95–17.41 (−0.64)	19.45–18.54 (−0.23)	16.87–18.16 (0.32)	15.16–18.16 (0.12)
Piauí	18.48–26.08 (1.90)	25.70–23.06 (−0.66)	21.55–18.93 (−0.66)	19.19–17.82 (−0.34)	17.28–18.58 (0.33)	18.48–18.58 (0.00)
Ceará	39.26–28.15 (−2.78)	28.58–18.86 (−2.43)	18.38–14.87 (−0.88)	16.03–15.29 (−0.19)	14.11–13.86 (−0.06)	39.26–13.86 (−1.06)
Rio Grande do Norte	23.31–24.11 (0.20)	20.76–18.37 (−0.60)	17.24–16.00 (−0.31)	16.56–14.54 (−0.51)	13.76–13.96 (0.05)	23.31–13.96 (−0.39)
Paraíba	61.57–26.34 (−8.81)	25.61–20.92 (−1.17)	19.43–16.45 (−0.74)	16.99–15.18 (−0.45)	13.60–17.57 (0.99)	61.57–17.57 (−1.83)
Pernambuco	57.58–29.97 (−6.90)	30.31–21.63 (−2.17)	19.67–16.47 (−0.80)	16.20–14.26 (−0.48)	14.51–15.78 (0.32)	57.58–15.78 (−1.74)
Alagoas	39.15–34.84 (−1.08)	31.92–25.49 (−1.61)	21.63–17.86 (−0.94)	18.39–15.94 (−0.61)	14.71–15.70 (0.25)	39.15–15.70 (−0.98)
Sergipe	31.64–32.95 (0.33)	29.99–21.84 (−2.04)	20.96–18.73 (−0.56)	17.15–17.21 (0.02)	18.89–20.37 (0.37)	31.64–20.37 (−0.47)
Bahia	36.33–26.32 (−2.50)	26.56–22.66 (−0.98)	21.38–19.34 (−0.51)	19.24–17.21 (−0.51)	17.16–17.89 (0.18)	36.33–17.89 (−0.77)
Southeast Region	**32.29–19.11 (−3.30)**	**19.00–16.13 (−0.72)**	**15.70–14.13 (−0.39)**	**13.82–13.10 (−0.18)**	**12.95–13.83 (0.22)**	**32.29–13.83 (−0.77)**
Minas Gerais	37.80–20.53 (−4.32)	20.31–17.44 (−0.72)	17.12–14.70 (−0.61)	14.08–13.36 (−0.18)	12.77–13.57 (0.20)	37.80–13.57 (−1.01)
Espírito Santo	30.47–18.76 (−2.93)	19.43–16.74 (−0.67)	16.80–13.74 (−0.76)	13.04–12.64 (−0.10)	12.39–13.51 (0.28)	30.47–13.51 (−0.71)
Rio de Janeiro	36.36–20.73 (−3.91)	20.56–17.11 (−0.86)	16.72–16.08 (−0.16)	15.15–14.47 (−0.17)	14.71–15.71 (0.25)	36.36–15.71 (−0.86)
São Paulo	28.59–17.88 (−2.68)	17.76–15.14 (−0.66)	14.61–13.22 (−0.35)	13.29–12.53 (−0.19)	12.44–13.33 (0.22)	28.59–13.33 (−0.64)
South Region	**27.14–18.64 (−2.12)**	**18.82–15.11 (−0.93)**	**14.77–12.90 (−0.47)**	**12.43–11.74 (−0.17)**	**11.55–12.12 (0.14)**	**27.14–12.12 (−0.63)**
Paraná	32.22–19.35 (−3.22)	19.68–15.50 (−1.04)	15.39–13.48 (−0.48)	12.73–11.79 (−0.24)	12.23–12.45 (0.06)	32.22–12.45 (−0.82)
Santa Catarina	23.39–18.03 (−1.34)	16.82–14.57 (−0.56)	13.61–12.37 (−0.31)	12.14–11.39 (−0.19)	10.77–11.54 (0.19)	23.39–11.54 (−0.49)
Rio Grande do Sul	23.87–18.23 (−1.41)	19.03–15.00 (−1.01)	14.81–12.60 (−0.55)	12.29–11.92 (−0.09)	11.35–12.20 (0.21)	23.87–12.20 (−0.49)
Central-West Region	**24.75–20.75 (−1.00)**	**20.07–17.85 (−0.56)**	**17.04–15.85 (−0.30)**	**16.11–13.76 (−0.59)**	**13.88–15.33 (0.36)**	**24.75–15.33 (−0.39)**
Mato Grosso do Sul	37.41–25.02 (−3.10)	25.22–23.28 (−0.49)	19.84–16.09 (−0.94)	16.03–12.65 (−0.85)	13.53–15.12 (0.40)	37.41–15.12 (−0.93)
Mato Grosso	19.04–24.41 (1.34)	22.58–19.71 (−0.72)	19.40–17.13 (−0.57)	18.04–15.45 (−0.65)	14.65–17.48 (0.71)	19.04–17.48 (−0.07)
Goiás	23.18–19.28 (−0.98)	18.71–16.70 (−0.50)	16.13–16.32 (0.05)	15.90–13.71 (−0.55)	14.62–15.30 (0.17)	23.18–15.30 (−0.33)
Federal District	22.80–16.24 (−1.64)	15.77–13.31 (−0.61)	13.54–13.13 (−0.10)	14.33–12.81 (−0.38)	11.54–12.19 (0.16)	22.80–12.19 (−0.44)

Table 1: **Under-five mortality rate (U5MR) and absolute annual change (AAC) by regions and states of Brazil, 1998–2022.**

Pernambuco (Northeast region) ($AAC_{1998–2022}$ = −1.51); and for NMR in Mato Grosso do Sul (Central-West region) ($AAC_{1998–2022}$ = −0.51). The lowest reductions in rates were observed in Amapá (North region) for all age groups: U5MR ($AAC_{1998–2022}$ = −0.05), IMR ($AAC_{1998–2022}$ = −0.01), and NMR ($AAC_{1998–2022}$ = −0.02).

During the first period (1998–2002), the most significant annual reduction in U5MR was observed in the Southeast and South regions. For North, Northeast and Central-West regions, the highest reduction occurred during the second period (2003–2007). Although U5MR rates declined across all states in the first four periods, all states (except Ceará in the Northeast region) experienced positive AAC in the most recent period (Table 1). Despite this, almost all states are expected to meet the UN goals for U5MR by 2030, but none are projected to achieve the IPEA goals. Moreover, two states, Roraima in the North region and Paraíba in the Northeast region, are unlikely to meet the SDG target for U5MR by 2030 if the final trend continues (Fig. 2).

IMR exhibited the greatest variability across regions (Table 2). The Southeast, South, and Central-West regions achieved the SDG target between 2003 and 2007, while the Northeast and North regions did not reach it until between 2013 and 2017. The Southeast region experienced the highest reduction in IMR during the early years of the study period ($AAC_{1998–2002}$ = −2.91). Although all regions met the SDG targets for IMR on average, four states (Maranhão, Paraíba, Bahia, and Mato Grosso) are projected not to meet this target by 2030 if current trends persist. Additionally, five states (Acre, Roraima, Amapá, Piauí, and Sergipe) are unlikely to achieve the SDG targets by 2030. No state is expected to meet the adjusted IPEA goal for IMR by 2030 (Fig. 2).

Regarding NMR, the South region achieved the SDG target before the start of the millennium. The Central-West and Southeast regions met this target between

Region/State	Infant mortality rate per 1000 live births (AAC)					
	1998–2002	2003–2007	2008–2012	2013–2017	2018–2022	1998–2022
Brazil	**27.54–19.18 (−2.09)**	**18.87–15.67 (−0.80)**	**15.02–13.46 (−0.39)**	**13.42–12.39 (−0.26)**	**12.18–12.59 (0.10)**	**27.54–12.59 (−0.62)**
North Region	**24.26–21.74 (−0.63)**	**21.41–18.49 (−0.73)**	**17.61–16.58 (−0.26)**	**16.48–15.42 (−0.26)**	**15.35–15.11 (−0.06)**	**24.26–15.11 (−0.38)**
Rondônia	24.06–21.55 (−0.63)	21.17–19.39 (−0.44)	16.27–13.73 (−0.64)	13.91–12.83 (−0.27)	12.71–13.41 (0.18)	24.06–13.41 (−0.44)
Acre	34.76–21.53 (−3.31)	20.33–22.06 (0.43)	17.89–16.41 (−0.37)	16.40–13.63 (−0.69)	16.50–17.19 (0.17)	33.67–17.19 (−0.73)
Amazonas	33.67–22.82 (−2.71)	21.50–16.97 (−1.13)	16.75–16.80 (0.01)	17.22–16.54 (−0.17)	16.05–15.69 (−0.09)	33.67–15.69 (−0.75)
Roraima	25.54–13.01 (−3.13)	15.85–17.12 (0.32)	16.63–16.51 (−0.03)	19.33–17.89 (−0.36)	19.93–18.79 (−0.29)	25.54–18.79 (−0.28)
Pará	21.48–21.88 (0.10)	22.30–18.74 (−0.89)	18.19–17.04 (−0.29)	16.46–15.39 (−0.27)	15.04–14.66 (−0.09)	21.48–14.66 (−0.28)
Amapá	18.34–24.16 (1.46)	20.73–20.80 (0.02)	22.64–20.41 (−0.56)	19.92–19.61 (−0.08)	18.53–18.07 (−0.12)	18.34–18.07 (−0.01)
Tocantins	13.59–20.10 (1.63)	19.69–17.47 (−0.55)	15.36–14.18 (−0.29)	13.60–12.39 (−0.30)	12.68–12.33 (−0.09)	13.59–12.33 (−0.05)
Northeast Region	**31.95–23.74 (−2.05)**	**23.17–18.36 (−1.20)**	**17.13–15.04 (−0.52)**	**15.48–14.06 (−0.36)**	**13.52–14.09 (0.14)**	**31.95–14.09 (−0.74)**
Maranhão	12.29–20.34 (2.01)	19.17–16.97 (−0.55)	16.45–14.68 (−0.44)	16.49–15.82 (−0.17)	14.07–15.32 (0.31)	12.29–15.32 (0.13)
Piauí	15.25–22.21 (1.74)	21.98–19.83 (−0.54)	18.44–16.62 (−0.46)	16.39–15.59 (−0.20)	14.85–15.76 (0.23)	15.25–15.76 (0.02)
Ceará	32.13–24.12 (−2.00)	24.56–16.11 (−2.11)	15.70–12.68 (−0.76)	13.82–13.20 (−0.16)	12.12–11.73 (−0.10)	32.13–11.73 (−0.85)
Rio Grande do Norte	18.80–21.11 (0.58)	17.40–15.73 (−0.42)	14.61–14.02 (−0.15)	14.42–12.31 (−0.53)	11.72–11.06 (−0.16)	18.80–11.06 (−0.32)
Paraíba	48.89–22.86 (−6.51)	22.02–18.25 (−0.94)	16.56–14.50 (−0.52)	14.56–13.29 (−0.32)	11.68–14.72 (0.76)	48.89–14.72 (−1.42)
Pernambuco	49.43–25.76 (−5.92)	26.07–18.69 (−1.85)	16.98–14.22 (−0.69)	14.12–12.12 (−0.50)	12.39–13.27 (0.22)	49.43–13.27 (−1.51)
Alagoas	33.15–30.19 (−0.74)	27.27–21.44 (−1.46)	18.60–15.16 (−0.86)	16.23–13.40 (−0.71)	12.53–12.83 (0.07)	33.15–12.83 (−0.85)
Sergipe	25.98–28.69 (0.68)	25.97–18.57 (−1.85)	17.74–16.27 (−0.37)	15.10–15.38 (0.07)	16.81–17.63 (0.20)	25.98–17.63 (−0.35)
Bahia	29.97–22.55 (−1.85)	23.01–19.75 (−0.81)	18.41–17.02 (−0.35)	17.03–15.10 (−0.48)	15.19–15.35 (0.04)	29.97–15.35 (−0.61)
Southeast Region	**28.06–16.41 (−2.91)**	**16.25–13.82 (−0.61)**	**13.51–12.24 (−0.32)**	**11.98–11.31 (−0.17)**	**11.17–11.64 (0.12)**	**28.06–11.64 (−0.68)**
Minas Gerais	32.65–17.72 (−3.73)	17.41–14.84 (−0.64)	14.70–12.72 (−0.50)	12.15–11.43 (−0.18)	10.96–11.37 (0.10)	32.65–11.37 (−0.89)
Espírito Santo	25.66–16.09 (−2.39)	16.36–13.86 (−0.63)	14.50–11.47 (−0.76)	11.04–10.67 (−0.09)	10.58–10.79 (0.05)	25.66–10.79 (−0.62)
Rio de Janeiro	31.41–17.89 (−3.38)	17.62–14.74 (−0.72)	14.39–13.81 (−0.14)	13.12–12.40 (−0.18)	12.66–13.16 (0.12)	31.41–13.16 (−0.76)
São Paulo	25.06–15.29 (−2.44)	15.18–13.04 (−0.54)	12.60–11.54 (−0.26)	11.57–10.92 (−0.16)	10.77–11.31 (0.13)	25.06–11.31 (−0.57)
South Region	**23.05–16.04 (−1.75)**	**15.78–12.94 (−0.71)**	**12.65–11.10 (−0.39)**	**10.70–10.15 (−0.14)**	**9.94–10.23 (0.07)**	**23.05–10.23 (−0.53)**
Paraná	27.33–16.82 (−2.63)	16.49–13.21 (−0.82)	13.09–11.66 (−0.36)	10.95–10.36 (−0.15)	10.33–10.32 (0.00)	27.33–10.32 (−0.71)
Santa Catarina	19.59–15.27 (−1.08)	14.10–12.77 (−0.33)	11.69–10.61 (−0.27)	10.46–9.93 (−0.13)	9.54–9.79 (0.06)	19.59–9.79 (−0.41)
Rio Grande do Sul	20.45–15.64 (−1.20)	15.97–12.75 (−0.80)	12.76–10.80 (−0.49)	10.57–10.07 (−0.13)	9.80–10.49 (0.17)	20.45–10.49 (−0.41)
Central-West Region	**20.77–17.33 (−0.86)**	**16.68–14.81 (−0.47)**	**14.48–13.59 (−0.22)**	**13.62–11.65 (−0.49)**	**11.79–12.56 (0.19)**	**17.33–12.56 (−0.34)**
Mato Grosso do Sul	32.59–20.33 (−3.06)	20.13–19.19 (−0.24)	16.54–13.37 (−0.79)	12.84–10.57 (−0.57)	11.32–12.35 (0.26)	32.59–12.35 (−0.84)
Mato Grosso	14.78–20.65 (1.47)	18.52–16.23 (−0.57)	16.11–13.97 (−0.53)	14.57–12.57 (−0.50)	12.14–14.08 (0.48)	14.78–14.08 (−0.03)
Goiás	19.53–16.18 (−0.84)	15.92–13.97 (−0.49)	13.88–14.40 (0.13)	13.86–11.87 (−0.50)	12.48–12.66 (0.04)	19.53–12.66 (−0.29)
Federal District	19.17–13.62 (−1.39)	13.30–11.09 (−0.55)	11.89–11.63 (−0.06)	12.73–11.08 (−0.41)	10.25–10.08 (−0.04)	19.17–10.08 (−0.38)

Table 2: **Infant mortality rate (IMR) and absolute annual change (AAC) by regions and states of Brazil, 1998–2022.**

2003 and 2007, while the North and Northeast regions reached it between 2008. The Southeast region demonstrated the highest reduction in NMR during the early years of the study period ($AAC_{1998–2002} = -1.37$), whereas other regions showed a faster reduction between 2003 and 2007. In the most recent period, only the North region continued to reduce NMR ($AAC_{2018–2022} = -0.23$), while other regions either stabilized or saw an increase in this rate (Table 3). Despite these variations, all regions are expected to meet the SDG targets for NMR by 2030, though none are projected to achieve the IPEA targets. Roraima (North) and Mato Grosso do Sul (Central-West) may meet the IPEA targets for NMR by 2030 if current trends continue. This is the only possibility of reaching the IPEA targets in Brazil (Fig. 2).

Mortality rate proportions among different age groups for children under five were also analyzed, as shown in Fig. 3. On average, there was an increase in the proportion of neonatal deaths and a decrease in the proportion of post-neonatal deaths until 2017. In the last period (2018–2022), the proportion of post-neonatal deaths increased. Throughout the study period, in all regions, early neonatal deaths accounted for the largest proportion of U5MR, while mortality among children aged 1–4 years accounted for the lowest proportion of deaths across all evaluated years. All regions reported an increased mortality in the post-neonatal stratum in the 2018–2022 period. The Northeast region had the highest proportion of early neonatal deaths, whereas the North and Central-West regions had the highest proportions of child deaths.

Discussion

Our comprehensive analysis of Brazilian childhood mortality rates from 1998 to 2022 reveals significant insights into regional variations and trends concerning

Region/State	Neonatal mortality rate per 1000 live births (AAC)					
	1998–2002	2003–2007	2008–2012	2013–2017	2018–2022	1998–2022
Brazil	**14.43–12.64 (−0.45)**	**12.27–10.66 (−0.40)**	**10.28–9.33 (−0.24)**	**9.20–8.76 (−0.11)**	**8.54–8.47 (−0.02)**	**14.43–8.47 (−0.25)**
North Region	**12.40–13.75 (0.34)**	**13.76–12.15 (−0.40)**	**11.62–10.82 (−0.20)**	**10.70–10.43 (−0.07)**	**10.40–9.50 (−0.23)**	**12.40–9.50 (−0.12)**
Rondônia	12.47–14.50 (0.51)	14.89–12.52 (−0.59)	10.79–8.94 (−0.46)	8.64–8.36 (−0.07)	8.15–8.63 (0.12)	12.47–8.63 (−0.16)
Acre	14.52–12.37 (−0.54)	12.26–13.41 (0.29)	10.28–9.46 (−0.21)	10.48–8.86 (−0.40)	10.70–9.11 (−0.40)	14.52–9.11 (−0.23)
Amazonas	15.97–13.68 (−0.57)	12.61–10.24 (−0.59)	10.21–10.29 (0.02)	10.25–10.68 (0.11)	10.69–9.40 (−0.32)	15.97–9.40 (−0.27)
Roraima	15.20–8.23 (−1.74)	8.56–10.44 (0.47)	9.90–9.34 (−0.14)	11.28–9.97 (−0.33)	13.11–9.32 (−0.95)	15.20–9.32 (−0.25)
Pará	11.54–13.99 (0.61)	14.78–12.65 (−0.53)	12.46–11.80 (−0.16)	11.24–11.07 (−0.04)	10.43–9.77 (−0.17)	11.54–9.77 (−0.07)
Amapá	12.04–17.54 (1.37)	15.51–17.95 (0.61)	17.68–14.64 (−0.76)	14.77–12.27 (−0.62)	12.80–11.68 (−0.28)	12.04–11.68 (−0.02)
Tocantins	6.47–12.60 (1.53)	11.90–10.88 (−0.26)	9.72–8.32 (−0.35)	8.67–8.42 (−0.06)	8.71–8.29 (−0.10)	6.47–8.29 (0.08)
Northeast Region	**12.41–14.84 (0.61)**	**14.27–12.66 (−0.40)**	**11.81–10.77 (−0.26)**	**10.95–10.10 (−0.21)**	**9.59–9.63 (0.01)**	**12.41–9.63 (−0.12)**
Maranhão	6.66–12.16 (1.38)	10.97–11.15 (0.05)	11.15–10.24 (−0.23)	11.75–11.21 (−0.13)	9.72–10.19 (0.12)	6.66–10.19 (0.15)
Piauí	8.94–14.52 (1.40)	14.34–14.41 (0.02)	13.50–12.28 (−0.31)	11.91–10.36 (−0.39)	10.26–9.82 (−0.11)	8.94–9.82 (0.04)
Ceará	8.50–15.05 (1.64)	15.30–10.81 (−1.12)	10.63–9.02 (−0.40)	9.95–9.19 (−0.19)	8.62–8.03 (−0.15)	8.50–8.03 (−0.02)
Rio Grande do Norte	7.54–13.92 (1.59)	11.58–10.75 (−0.21)	10.36–10.15 (−0.05)	9.87–8.50 (−0.34)	8.46–7.77 (−0.17)	7.54–7.77 (0.01)
Paraíba	15.19–14.30 (−0.22)	13.82–13.14 (−0.17)	11.26–10.66 (−0.15)	9.99–9.55 (−0.11)	8.01–9.98 (0.49)	15.19–9.98 (−0.22)
Pernambuco	20.01–14.94 (−1.27)	14.85–12.08 (−0.69)	11.05–9.97 (−0.27)	9.70–8.67 (−0.26)	8.71–8.91 (0.05)	20.01–8.91 (−0.46)
Alagoas	10.73–16.74 (1.50)	13.90–13.78 (−0.03)	12.23–9.77 (−0.61)	10.44–8.89 (−0.39)	8.65–8.61 (−0.01)	10.73–8.61 (−0.09)
Sergipe	13.32–19.93 (1.65)	16.81–12.90 (−0.98)	12.78–11.38 (−0.35)	10.49–11.84 (0.34)	12.70–11.85 (−0.21)	13.32–11.85 (−0.06)
Bahia	13.05–15.10 (0.51)	15.46–14.57 (−0.22)	13.21–12.65 (−0.14)	12.51–11.47 (−0.26)	11.02–11.03 (0.00)	13.05–11.03 (−0.08)
Southeast Region	**17.00–11.53 (−1.37)**	**11.14–9.44 (−0.42)**	**9.30–8.38 (−0.23)**	**8.18–7.96 (−0.05)**	**7.74–7.91 (0.04)**	**17.00–7.91 (−0.38)**
Minas Gerais	18.75–12.59 (−1.54)	12.1 2–10.39 (−0.43)	10.49–8.94 (−0.39)	8.61–8.15 (−0.11)	7.94–8.07 (0.03)	18.75–8.07 (−0.44)
Espírito Santo	14.87–11.67 (−0.80)	10.93–9.43 (−0.38)	10.12–7.82 (−0.58)	7.53–7.70 (0.04)	7.39–7.17 (−0.05)	14.87–7.17 (−0.32)
Rio de Janeiro	18.70–12.66 (−1.51)	12.23–10.02 (−0.55)	9.62–9.10 (−0.13)	8.61–8.57 (−0.01)	8.40–8.48 (0.02)	18.70–8.48 (−0.43)
São Paulo	15.80–10.61 (−1.30)	10.29–8.82 (−0.37)	8.60–7.93 (−0.17)	7.89–7.68 (−0.05)	7.44–7.70 (0.06)	15.80–7.70 (−0.34)
South Region	**13.19–10.48 (−0.68)**	**10.16–8.70 (−0.36)**	**8.78–7.71 (−0.27)**	**7.26–7.28 (0.01)**	**7.23–7.10 (−0.03)**	**13.19–7.10 (−0.25)**
Paraná	15.45–11.55 (−0.98)	10.98–9.09 (−0.47)	9.20–8.18 (−0.26)	7.60–7.49 (−0.03)	7.55–7.10 (−0.11)	15.45–7.10 (−0.35)
Santa Catarina	10.94–9.93 (−0.25)	9.25–8.64 (−0.15)	8.10–7.58 (−0.13)	6.99–7.38 (0.10)	6.91–6.83 (−0.02)	10.94–6.83 (−0.17)
Rio Grande do Sul	12.04–9.64 (−0.60)	9.79–8.30 (−0.37)	8.73–7.28 (−0.36)	7.05–6.97 (−0.02)	7.10–7.32 (0.05)	12.04–7.32 (−0.20)
Central-West Region	**12.39–11.90 (−0.12)**	**11.53–9.99 (−0.39)**	**9.76–9.57 (−0.05)**	**9.31–8.32 (−0.25)**	**8.35–8.18 (−0.04)**	**12.39–8.18 (−0.18)**
Mato Grosso do Sul	18.97–12.95 (−1.50)	12.69–12.25 (−0.11)	11.30–8.69 (−0.65)	8.04–7.75 (−0.07)	7.61–6.84 (−0.19)	18.97–6.84 (−0.51)
Mato Grosso	7.14–14.30 (1.79)	12.53–10.81 (−0.43)	10.58–9.70 (−0.22)	9.71–8.59 (−0.28)	8.06–9.21 (0.29)	7.14–9.21 (0.09)
Goiás	12.01–11.32 (−0.17)	11.50–9.61 (−0.47)	9.50–10.39 (0.22)	9.67–8.47 (−0.30)	9.04–8.50 (−0.14)	12.01–8.50 (−0.15)
Federal District	12.83–9.69 (−0.78)	9.57–7.89 (−0.42)	7.90–8.51 (0.15)	9.27–8.19 (−0.27)	7.92–7.24 (−0.17)	12.83–7.24 (−0.23)

Table 3: Neonatal mortality rate (NMR) and absolute annual change (AAC) by regions and states of Brazil, 1998–2022.

both UN SDGs and IPEA targets. This study is, to the best of our knowledge, the first of its scale for Brazil. We assessed regional mortality patterns and age-specific components to identify areas requiring more focused interventions and to highlight which age groups are at the highest risk. Overall, Brazil has made notable progress in reducing mortality rates, particularly in the first decade of the millennium, with a marked decrease in regional disparities and concentration of deaths in the neonatal period. Although the country is on track to meet the SDG targets for NMR, IMR, and U5MR by 2030, it is unlikely to achieve the IPEA targets. Specifically, only two states have a realistic chance of meeting the IPEA NMR target, while 11 states are projected to fall short of the UN goals for IMR and U5MR rates by 2030.

The latest analysis by the UN inter-agency group on childhood mortality reveals that 122 countries have already met the SDG target for U5MR, and 22 more are expected to achieve this target by 2030. However, 69 countries are projected to miss the SDG target for NMR by 2030.[3] Our study indicates that, like other countries, Brazil exhibited a more pronounced decline in mortality rates during the early 2000s, with a deceleration in progress during the last decade (2010–2019). Globally, the decrease in NMR has been slower compared to U5MR, leading to a growing proportion of neonatal deaths within the overall U5MR over time.[3,12]

A similar study conducted in India highlighted a shift towards higher mortality rates in the early neonatal period, with reduced rates in the post-neonatal stratum and among children aged 1 to 4. This study also noted a more substantial decline in deaths among children older than one year and a reduction in regional disparities. Despite these improvements, India is unlikely to meet the targets for early neonatal and post-neonatal mortality by 2030.[14]

Child mortality rates are highly responsive to basic public health interventions such as vaccination,

	Neonatal (SDG target 12.0) (IPEA target 5.3) Rate (AAC$_{2018\text{-}2022}$)	Infant (SDG target 15.7) (IPEA target 7.7) Rate (AAC$_{2018\text{-}2022}$)	Under-5 (SDG target 25.0) (IPEA target 8.3) Rate (AAC$_{2018\text{-}2022}$)
Brazil	**8 47 (-0 02)**	**12.59 (0.10)**	**15.04 (0.22)**
North Region	**9 50 (-0 23)**	**15.11 (-0.06)**	**18.52 (0.11)**
Rondônia	8.63 (0.12)	13.41 (0.18)	16.18 (0.47)
Acre	9.11 (-0.40)	17.19 (0.17)	20.64 (0.30)
Amazonas	9.40 (-0.32)	15.69 (-0.09)	19.46 (0.03)
Roraima	9.32 (-0.95)	18.79 (-0.29)	24.60 (0.15)
Pará	9.77 (-0.17)	14.66 (-0.09)	17.76 (0.05)
Amapá	11.68 (-0.28)	18.07 (-0.12)	21.81 (0.19)
Tocantins	8.29 (-0.10)	12.33 (-0.09)	15.48 (0.05)
Northeast Region	**9.63 (0.01)**	**14.09 (0.14)**	**16.69 (0.25)**
Maranhão	10.19 (0.12)	15.32 (0.31)	18.16 (0.32)
Piauí	9.82 (-0.11)	15.76 (0.23)	18.58 (0.33)
Ceará	8.03 (-0.15)	11.73 (-0.10)	13.86 (-0.06)
Rio Grande do Norte	7.77 (-0.17)	11.06 (-0.16)	13.96 (0.05)
Paraíba	9.98 (0.49)	14.72 (0.76)	17.57 (0.99)
Pernambuco	8.91 (0.05)	13.27 (0.22)	15.78 (0.32)
Alagoas	8.61 (-0.01)	12.83 (0.07)	15.70 (0.25)
Sergipe	11.85 (-0.21)	17.63 (0.20)	20.37 (0.37)
Bahia	11.03 (0.00)	15.35 (0.04)	17.89 (0.18)
Southeast Region	**7.91 (0.04)**	**11.64 (0.12)**	**13.83 (0.22)**
Minas Gerais	8.07 (0.03)	11.37 (0.10)	13.57 (0.20)
Espírito Santo	7.17 (-0.05)	10.79 (0.05)	13.51 (0.28)
Rio de Janeiro	8.48 (0.02)	13.16 (0.12)	15.71 (0.25)
São Paulo	7.70 (0.06)	11.31 (0.13)	13.33 (0.22)
South Region	**7.10 (-0.03)**	**10.23 (0.07)**	**12.12 (0.14)**
Paraná	7.10 (-0.11)	10.32 (0.00)	12.45 (0.06)
Santa Catarina	6.83 (-0.02)	9.79 (0.06)	11.54 (0.19)
Rio Grande do Sul	7.32 (0.05)	10.49 (0.17)	12.20 (0.21)
Central-West Region	**8.18 (-0.04)**	**12.56 (0.19)**	**15.33 (0.36)**
Mato Grosso do Sul	6.84 (-0.19)	12.35 (0.26)	15.12 (0.40)
Mato Grosso	9.21 (0.29)	14.08 (0.48)	17.48 (0.71)
Goiás	8.50 (-0.14)	12.66 (0.04)	15.30 (0.17)
Distrito Federal	7.24 (-0.17)	10.08 (-0.04)	12.19 (0.16)

Met the SDG target by 2022 and will continue to meet it in 2030; did not meet the IPEA target by 2022 but will meet it in 2030

Met the SDG target by 2022 and will continue to meet it in 2030; did not meet the IPEA target by 2022 and will not meet it in 2030

Met the SDG target by 2022 but will not meet it in 2030

Did not meet the SDG target by 2022 and will not meet it in 2030

Fig. 2: Distribution of Brazilian states' status for early-life mortality 2030 targets, 2022. Legends/Abbreviations: SDG = Sustainable Development Goals, IPEA = Instituto de Pesquisa Econômica Aplicada (Applied Economic Research).

sanitation, and nutrition, topics in which Brazil has made notable progress in recent decades.[8] Brazil's primary care system, one of the largest globally, expanded its coverage by over 200% from 1998 to 2019.[15] This extensive coverage is a crucial factor in reducing child mortality.[15–17] However, from 2018 to 2022, U5MRs increased in Brazil, accompanied by a higher proportion of post-neonatal deaths. This trend may reflect a decline in the quality of primary care services, as effective child health interventions are closely linked to high-quality primary care.

Significant shortcomings in respect of Brazil's childcare services have been observed, with regional disparities that suggest poorer pediatric care in high-mortality areas.[18] Studies comparing states in the Northeast and South found that children from the Northeast are less likely to receive adequate care. Additionally, the quality-of-care is generally higher in the South and Southeast regions, as shown by a study on newborn care during the first week of life.[19–21]

National dietary surveys reveal a noteworthy pattern: children living in more developed urban areas of Brazil, especially in the South and Southeast regions, tend to have healthier diets. This trend is closely linked to greater access to supplementary pediatric care, highlighting how urban advantages and additional

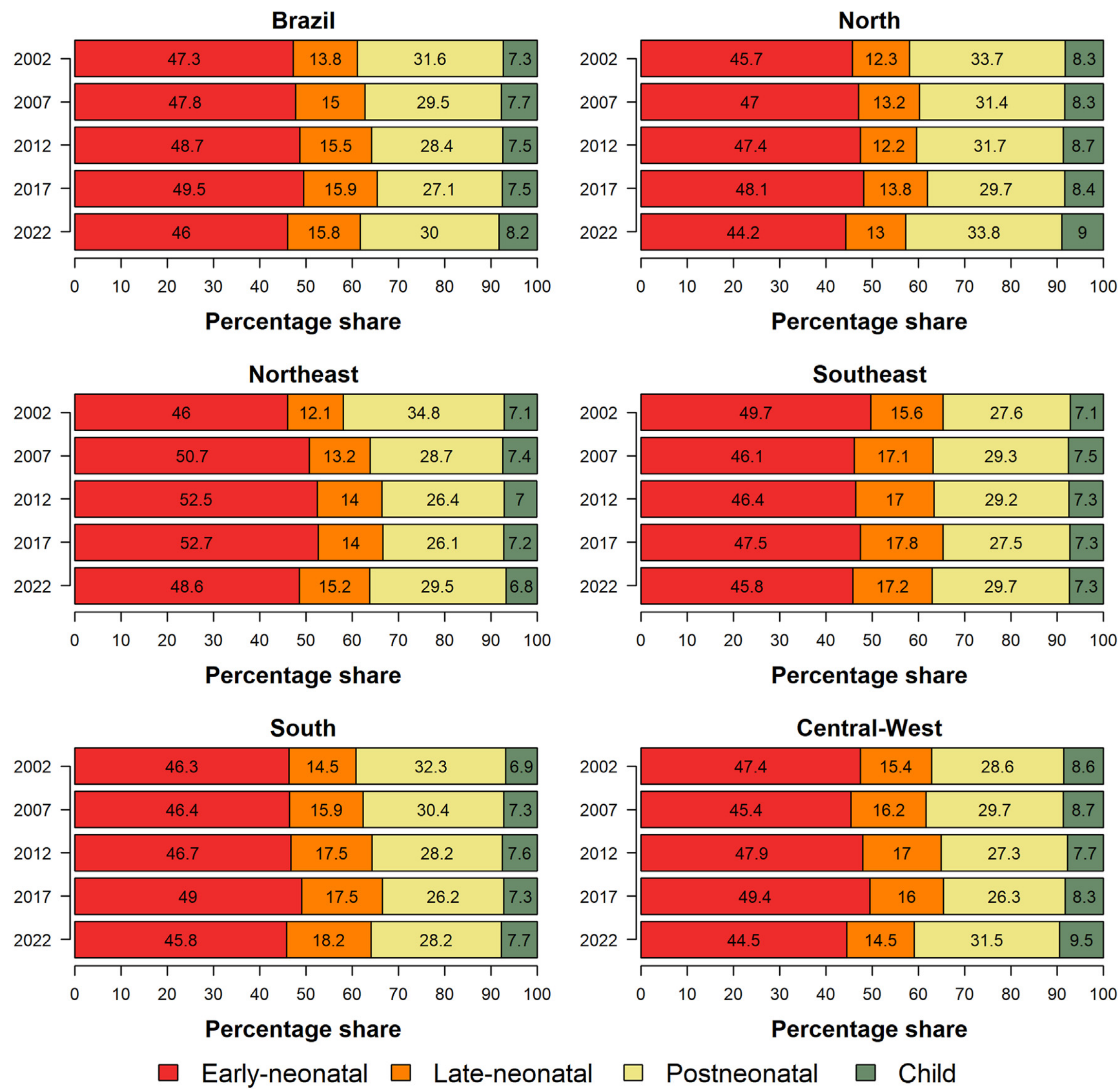

Fig. 3: Age group proportions in under-5 mortality rate across Brazilian regions, 1998–2022.

healthcare support contribute to better outcomes for children.[22]

It is important to consider that the last period studied (2018–2022) encompassed the peak of the COVID-19 pandemic. While the 2023 UN report suggests that the pandemic had no significant impact on global child mortality trends, data indicates that Brazil experienced higher COVID-19-related child mortality compared to other countries, particularly in less developed regions.[23,24] Moreover, many regions in Brazil and globally faced substantial disruptions to essential child health services during the pandemic and limitations on fundamental child health services in primary care, which may have contributed to the increased mortality rates from preventable causes during and in the years following the pandemic.[25–28]

Reducing neonatal mortality poses a significant challenge, as it necessitates comprehensive interventions across multiple levels. Effective strategies include prenatal care, skilled birth assistance, immediate newborn care, and specialized support for premature and ill infants.[29–31] The *Nascer no Brasil* survey, a comprehensive study of labor and births in Brazil, revealed that essential perinatal care practices are inadequately provided nationwide. Key interventions such as skin-to-skin contact, breastfeeding within the first hour, antenatal corticosteroids, and the use of partograms were implemented in less than half of the country's births, with notable regional disparities.[32]

According to Lawn et al. (2023), broadening access to specialized neonatal care, such as neonatal intensive care units (NICUs), continuous positive airway pressure therapy, and therapeutic hypothermia, has the potential to prevent around 700,000 neonatal deaths each year.[33] Data from the *Nascer no Brasil* survey reveals that fewer than half of high-risk births during the study period took place in facilities equipped with NICUs, with the lowest availability found in the North and Northeast regions.[34]

Because we observed that neonatal deaths account for the majority of the U5MR, we analyzed the structural adequacy of neonatal care and compared it between Brazilian states. The Brazilian Society of Pediatrics recommends an average of four NICU beds per 1000 live births and suggests adjusting this availability based on regional neonatal mortality rates and causes.[35] However, according to the National Registry of Health Establishments,[36] only nine states meet this recommendation, with seven located in the South or Southeast. In contrast, all ten states with the fewest NICU beds are in the North or Northeast regions (Supplementary Table S1). Even in states where the recommendation is met, this happens thanks to the complementary health service. Although most of the population relies on the public health system in Brazil,[8] only 50% of the neonatal beds are in the public sector. This disparity underscores the urgent need to improve NICU access in underserved areas to effectively address neonatal mortality.

According to the latest United Nations Development Program report (2022), Brazil is classified as an "upper-middle-income" country.[9] In 2022, countries also considered as "upper-middle-income" had an average U5MR of 13.3 per 1000 live births and an NMR of 6.9 per 1000 live births. Despite Brazil achieving the SDG targets, its rates are higher than the average for similar countries, with a U5MR of 15.1 per 1000 live births and an NMR of 8.5 per 1000 live births in 2022. The UN encourages nations that have met SDG targets to strive for the performance levels of higher-income countries, which have U5MRs around 5.0 and NMR of 2.9 per 1000 live births.[2,3] To reduce Brazil's NMR to around 5.0 across all regions and states, it is essential to enhance the availability of NICU beds and ensure equitable access to qualified healthcare professionals.[33]

This is a population-based ecological study, that relied on secondary data to analyze disaggregated patterns of early-life mortality over 25 years, which may present limitations related to the completeness of certain variables. However, the databases used here have shown high coverage and reliability regarding the total number of infant deaths and live births, particularly in more recent years. Despite regional variations—especially in the North and Northeast—coverage has improved over time, and national evaluations report completeness rates above 90% for all Brazilian regions. Furthermore, as with all ecological studies, our findings are subject to the possibility of ecological fallacy, since associations observed at the group level may not reflect individual-level relationships. Additionally, this study did not assess the ethnicity and sex of the child. We recognize that ethnicity could be an important disaggregated analysis, but due to limitations of the dataset, this analysis was not possible. Regarding sex, mortality rates between boys and girls in early age groups have been very similar in Brazil. Nonetheless, ecological designs remain valuable tools for monitoring trends in morbidity and mortality, and for assessing the population-level impact of public health policies over time.

This study has significant strengths and is notable as the first, to the best of our knowledge, to investigate disaggregated patterns of early-life mortality across Brazilian states over 25 years, and provides a comprehensive and detailed view of regional variations and progress towards SDG and IPEA targets. The extensive analysis over an extended period allows for the identification of trends and regional disparities that might be missed in shorter studies. The study's findings offer a solid foundation for future research and policymaking, highlighting the need for more tailored regional approaches and targeted strategies to meet sustainable development goals and improve early-life mortality outcomes. Continued analysis of these trends can enhance understanding of regional disparities and contribute to the development of more effective public policies in Brazil.

In conclusion, Brazilian children face significant disparities in survival chances based on their place of birth and residence. The variations in early-life mortality rates across different regions of Brazil underscore the urgent need to address these inequalities. To ensure the survival and well-being of Brazilian children, expanding healthcare coverage is essential, but it must be accompanied by a strong commitment to equity and the quality of care.[37,38] Efforts must be directed towards identifying and addressing the primary causes of mortality, particularly in regions with critical needs and among vulnerable age groups.

It is essential to enhance the quality of healthcare services at the primary level, ensuring that they are accessible and effective across the entire life course. This involves not only improving the availability of health services, but also ensuring that these services are equitable and meet the highest standards of care. Addressing these disparities requires a concerted effort to target the most significant risk factors and to implement strategies that are tailored to the specific needs of different regions and populations. By prioritizing both coverage and quality, Brazil can make meaningful progress in reducing early-life mortality and improving health outcomes for all Brazilian children.

Contributors
ASAL, DFL, VSS and RQG contributed to conceptualization and study design. ASAL, ELC, LBS and JRSS contributed to data acquisition. JRSS, ASAL and VSS contributed to data analysis and interpretation. ASAL drafted the manuscript. DFL, VSS and RQG contributed to the critical

revision of the manuscript. RQG and VSS provided administrative, technical, or material support, as well as supervision and mentorship. Each author contributed important intellectual content during manuscript drafting or revision and accepts accountability for the overall work by ensuring that questions pertaining to the accuracy or integrity of any portion of the work are appropriately investigated and resolved. ASAL and RQG take responsibility for the fact that this study has been reported honestly, accurately, and transparently; that no important aspects of the study have been omitted, and that any discrepancies from the study as planned have been explained. JRSS and ASAL had full access to all the data in the study, and have verified the accuracy of all underlying data. All authors had final responsibility for the decision to submit the manuscript for publication.

Data sharing statement

The data used are public and available from DATASUS or can be obtained by contacting the corresponding author.

Declaration of interests

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.lana.2025.101133.

References

1 United Nations (UN). *The millennium development goals report*; 2015. https://www.un.org/millenniumgoals. Accessed December 27, 2023.
2 United Nations (UN). *Levels & trends in child mortality: report 2023, estimates developed by the united nations interagency group for child mortality estimation*. New York, USA: UN; 2024.
3 Sharrow D, Hug L, You D, et al. Global, regional, and national trends in under-5 mortality between 1990 and 2019 with scenario-based projections until 2030: a systematic analysis by the UN Inter-agency Group for Child Mortality Estimation. *Lancet Glob Health*. 2022;10(2):e195–e206.
4 Brasil, Sistema de Informações em Saúde, Estatísticas Vitais. *DATASUS, Departamento de Informática do SUS*; 2022. https://datasus.saude.gov.br/informacoes-de-saude-tabnet/. Accessed May 7, 2024.
5 Instituto de Pesquisa Econômica Aplicada (IPEA). *Objetivos de Desenvolvimento do Milênio - Relatório Nacional de Acompanhamento*. Brasília, Brazil: IPEA; 2014.
6 Instituto Brasileiro de Geografia e Estatística (IBGE). *Estimativa da População*; 2023. https://www.ibge.gov.br/estatisticas/sociais/populacao/9103-estimativas-de-populacao.html. Accessed July 4, 2024.
7 Instituto de Pesquisa e Estatística Aplicada (IPEA). *PNUD, atlas do desenvolvimento humano no Brasil*; 2022. http://www.atlasbrasil.org.br/ranking/. Accessed July 4, 2024.
8 Paim J, Travassos C, Almeida C, Bahia L, Macinko J. Series - the Brazilian health system: history, advances, and challenges. *Lancet*. 2011;377:1778–1797.
9 United Nations Development Programme (UNDP). *Human development report: 2021/2022*. NewYork, USA: UNDP; 2022.
10 Paulson KR, Kamath AM, Alam T, et al. Global, regional, and national progress towards Sustainable Development Goal 3.2 for neonatal and child health: all-cause and cause-specific mortality findings from the Global Burden of Disease Study 2019. *Lancet*. 2021;398(10303):870–905.
11 Burstein R, Henry NJ, Collison ML, et al. Mapping 123 million neonatal, infant and child deaths between 2000 and 2017. *Nature*. 2019;574(7778):353–358.
12 Rosa-Mangeret F, Benski AC, Golaz A, et al. 2.5 million annual deaths—are neonates in low-and middle-income countries too small to Be seen? A bottom-up overview on neonatal morbi-mortality. *Trop Med Infect Dis*. 2022;64(7).
13 Lansky S, Friche AAL, Silva AAM, et al. Birth in Brazil survey: neonatal mortality profile, and maternal and child care. *Cad Saude Publ*. 2014;30(SUPPL1):S192–S207.
14 Subramanian SV, Kumar A, Pullum TW, Ambade M, Rajpal S, Kim R. Early-neonatal, late-neonatal, postneonatal, and child mortality rates across India, 1993–2021. *JAMA Netw Open*. 2024;7(5):E2410046.
15 Moncayo AL, Cavalcanti DM, Ordoñez JA, et al. Can primary health care mitigate the effects of economic crises on child health in Latin America? An integrated multicountry evaluation and forecasting analysis. *Lancet Glob Health*. 2024;12(6):e938–e946.
16 Sanhueza A, Cueva DA, Mujica OJ, Soliz P, Duran P. Income inequality as a determinant of neonatal mortality in the Americas during 2000–2019: implications for the attainment of Sustainable Development Goal target 3.2. *Int J Equity Health*. 2024;23(109).
17 Braz SGC, Raiher AP. Reducing child mortality and meeting millennium development goal 4 in Brazil. *Econ Soc Territ*. 2022;22(68):265–295.
18 Pedraza DF. Consulta de puericultura na Estratégia Saúde da Família em municípios do interior do estado da Paraíba, Brasil. *Cien Saude Colet*. 2023;28(8):2291–2302.
19 Oscar MCB, Simão DAS, Ribeiro GC, Vieira EWR. Neonatal visits in the first week of life in primary care: low prevalence and related factors. *Rev Bras Enferm*. 2022;75(4).
20 Santos AS, Duro SMS, Cade NV, Fachini LA, Tomasi E. Acesso ao atendimento de puericultura nas Regiões Nordeste e Sul do Brasil. *Rev Bras Saude Mater Infant*. 2017;17(3):447–460.
21 Domingues RMSM, Dias BAS, Azevedo BSD, et al. Use of outpatient health services by postpartum women and newborns: data from the Birth in Brazil study. *Cad Saude Publica*. 2020;36(5):e00119519.
22 Carvalho RBN, Louzada MLC, Rauber F, Levy RB. Characteristics associated with dietary patterns in Brazilian children under two years of age. *Rev Saude Publica*. 2023;56:118.
23 Lopes ASA, Vieira SCF, Porto RLS, et al. Coronavirus disease-19 deaths among children and adolescents in an area of Northeast, Brazil: why so many? *Trop Med Int Health*. 2021;26(1):115–119.
24 Santos VS, Siqueira TS, Atienzar AIC, et al. Spatial clusters, social determinants of health and risk of COVID-19 mortality in Brazilian children and adolescents: a nationwide population-based ecological study. *Lancet Reg Health Am*. 2022;13:e100311.
25 Somekh I, Somech R, Pettoello-Mantovani M, Somekh E. Changes in routine pediatric practice in light of coronavirus 2019 (COVID-19). *J Pediatr*. 2020;224:190–193.
26 Shibukawa BMC, Uema RTB, Piran CMG, et al. Repercussions of the pandemic of COVID-19: care of the pediatric population in Primary Health Care. *Rev Rene*. 2022;23:e72798.
27 Cabral IE, Pestana-Santos M, Ciuffo LL, Nunes YDR, Lomba MLLF. Child health vulnerabilities during the covid-19 pandemic in Brazil and Portugal. *Rev Lat Am Enfermagem*. 2021;29:e3422.
28 Takemoto MLS, Menezes MO, Andreucci CB, et al. Maternal mortality and COVID-19. *J Matern Fetal Neonatal Med*. 2020;35(12):2355–2361.
29 World Health Organization (WHO). *United Nations Children's Fund (UNICEF) Every Newborn: an action plan to end preventable deaths*. Geneva, Switzerland: WHO; 2014.
30 Leal MDC, Szwarcwald CL, Almeida PVB, et al. Reproductive, maternal, neonatal and child health in the 30 years since the creation of the Unified Health System (SUS). *Ciênc Saúde Coletiva*. 2018;23(6):1915–1928.
31 Bhutta ZA, Das JK, Bahl R, et al. Can available interventions end preventable deaths in mothers, newborn babies, and stillbirths, and at what cost? *Lancet*. 2014;384:347–370.
32 Menezes MAS, Gurgel R, Bittencourt SDA, Pacheco VE, Cipolotti R, Leal MDC. Health facility structure and maternal characteristics related to essential newborn care in Brazil: a cross-sectional study. *BMJ Open*. 2018;8(12).
33 Lawn JE, Bhutta ZA, Ezeaka C, Saugstad O. Ending preventable neonatal deaths: multicountry evidence to inform accelerated progress to the sustainable development goal by 2030. *Neonatology*. 2023;120:491–499.
34 Bittencourt SDA, Domingues RMSM, Costa RLG, Ramos MM, Leal MDC. Adequacy of public maternal care services in Brazil. *Reprod Health*. 2016;13(Suppl 1):256–265.

35 Sociedade Brasileira de Pediatria (SBP). *Relação do número de leitos de UTI neonatal por 1000 nascidos vivos - Estimativa da necessidade de leitos de UTI neonatal no Brasil*. Brazil: Rio de Janeiro; 2012.
36 Brasil. DATASUS. *Cadastro Nacional de Estabelecimentos de Saúde*; 2022. https://cnes.datasus.gov.br/pages/estabelecimentos/consulta.jsp. Accessed July 4, 2024.
37 World Health Organization (WHO). *Standards for improving quality of maternal and newborn care in health facilities*. Geneva, Switzerland: WHO; 2016.
38 Chou VB, Walker N, Kanyangarara M. Estimating the global impact of poor quality of care on maternal and neonatal outcomes in 81 low- and middle-income countries: a modeling study. *PLoS Med*. 2019;16(12).